



Therapeutic effects of Gunryeong-tang on the cardio-renal axis in an animal model of pulmonary hypertension

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ABSTRACT

Background: In the realm of cardiovascular and renal health, the modulation of oxidative stress and inflammation within the cardio-renal axis plays a pivotal role. This study investigates the therapeutic potential of Gunryeong-tang (GRT), a traditional Korean herbal formula combining Oryeongsan (Wulingsan) and Sagunza-tang (Sijunzi-tang), specifically in addressing cardio-renal axis dysfunction induced by pulmonary arterial hypertension (PAH-r). **Methods:** PAH-r models were induced through intraperitoneal injection of monocrotaline (MCT) at a daily dose of 50 mg/kg for 5 days to simulate pulmonary hypertension, a common cause of cardio-renal syndrome. The primary focus is on evaluating its anti-fibroinflammatory effects and its ability to modulate oxidative stress and inflammation. The treatment group received oral administration of GRT (100 mg/kg/day). **Results:** Various parameters were assessed, including right ventricular (RV) enlargement and fibrosis, lung weight, pulmonary fibrosis, expression of cardiac hypertrophy markers, and involvement of specific pathways (TGF- β /p-Smad2 and HMGB-1/TLR4/MyD88/NF- κ B). Renal dysfunction markers, such as blood urea nitrogen (BUN), creatinine clearance rate (Ccr), glomerular dilatation, tubular fibrosis, and protein expression of kidney injury molecule-1 (KIM-1) and neutrophil gelatinase-associated lipocalin (NGAL), were evaluated to highlight the formula's anti-fibroinflammatory effects. Treatment resulted in significant improvements in the PAH-r group. It alleviated RV enlargement and fibrosis, suppressed lung weight increase, and mitigated pulmonary fibrosis. The administration downregulated cardiac hypertrophy markers (CTGF, TGF- β , and α -SMA mRNA levels) and inhibited the TGF- β /p-Smad2 and HMGB-1/TLR4/MyD88/NF- κ B pathways, showcasing GRT's anti-fibroinflammatory properties. Additionally, the treatment ameliorated renal dysfunction, as evidenced by reduced BUN and Ccr levels, improved glomerular dilatation, and reduced protein expression of KIM-1 and NGAL. GRT demonstrates significant therapeutic potential in managing cardio-renal axis dysfunction induced by PAH-r, emphasizing its multifaceted benefits in modulating oxidative stress and inflammation within this axis. **Conclusions:** This study provides valuable insights into the underlying mechanisms, highlighting the modulation of cardiac hypertrophy markers and specific pathways. These findings underscore the promising role of GRT as a viable treatment option for cardio-renal axis dysfunction associated with PAH-r.

1. Introduction

Pulmonary arterial hypertension (PAH) is a debilitating disease characterized by an elevation in pulmonary artery pressure exceeding 25 mmHg. It primarily arises from progressive dysfunction in the pulmonary vasculature, resulting in increased resistance and pressures within the pulmonary arteries. These changes contribute to right ventricular (RV) hypertrophy, heart failure, and increased mortality.^{1,2} The

pathophysiology of PAH involves remodeling and constriction of pulmonary vessels, leading to restricted blood flow through the pulmonary arteries. Consequently, the RV faces heightened afterload, which triggers compensatory mechanisms including increased contractility and adaptation to changes in volume load.³ These adaptations ultimately lead to RV hypertrophy.⁴ In PAH, the RV undergoes concentric hypertrophy, deviating from its normal smaller size in comparison to the left ventricle, due to elevated systolic and diastolic pressures.⁵ Moreover,

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the elevated pressures in the RV contribute to the growth of muscle mass through protein synthesis and cardiomyocyte hypertrophy.⁶ Overall, PAH is characterized by increased pulmonary vascular resistance, RV hypertrophy, and subsequent heart failure. The remodeling and constriction of pulmonary vessels, along with elevated RV pressures, play crucial roles in the development and progression of PAH.

Cardio-renal axis refers to the complex interaction between cardiac and renal dysfunction, leading to diuretic disorders and exacerbation of heart failure.^{7,8} In this study, we specifically focus on understanding the intricate relationship between the lungs, right heart, and kidneys, as they play crucial roles in the multi-organ dysfunction observed in PAH. The administration of monocrotaline (MCT) induces an increase in atrial natriuretic peptide (ANP), brain natriuretic peptide (BNP), and β -myosin heavy chain (β -MHC) levels in the right ventricle, which serve as markers for cardiac hypertrophy.⁹ ANP directly promotes fibrotic effects, particularly on fibroblasts, by stimulating the expression of fibrosis-related genes like collagen I, collagen III, and connective tissue growth factor (CTGF) through upregulation of transforming growth factor- β (TGF- β).^{8,10} TGF- β plays a crucial role in regulating cellular activities, including vascular fibroblast and smooth muscle cell differentiation, proliferation, and migration. Upon binding to its cell membrane receptor, TGF- β activates downstream signaling molecules, such as mothers against decapentaplegic homolog 2/3 (Smad 2/3), which, upon phosphorylation, form a complex with Smad 4 and translocate to the nucleus for gene regulation.^{11,12} The TGF- β family is involved in various physiological processes and has been implicated in the pathogenesis of hypertension, atherosclerosis, PAH, and heart failure.¹³

Recent studies have highlighted the role of high mobility group box-1 (HMGB-1) in the development of PAH, as evidenced by animal models and cell experiments.^{14–16} HMGB-1 interacts with various membrane receptors, including toll-like receptor 4 (TLR4).¹⁷ TLR4 recognizes and binds extracellular HMGB-1, subsequently activating the myeloid differentiation factor 88 (MyD88) pathway and the nuclear factor-kappa B (NF- κ B) signaling pathway.¹⁸ NF- κ B is a critical regulator of innate and adaptive immunity, activating genes associated with inflammation, leukocyte migration, and proliferation.¹⁹ NF- κ B, along with cytokines such as tumor necrosis factor- α (TNF- α), interleukin-1 β (IL-1 β), and interleukin-6 (IL-6), plays a significant role in the pathogenesis of PAH.²⁰ Consequently, inhibiting NF- κ B is considered a potential therapeutic target for mitigating PAH.^{21,22} Progressive renal disease involves renal fibrosis and the restoration of normal renal structure.²³ MCT administration induces renal fibrosis and renal toxicity, leading to renal dysfunction and eventual renal failure.²⁴ Biomarkers such as neutrophil gelatinase-associated lipocalin (NGAL) and kidney injury molecule-1 (KIM-1) are produced by renal tubules after injury and serve as indicators for monitoring renal tubular damage.²⁴ This study aims to investigate the potential protective effects of Korean herbal formula (Gunryeong-tang, GRT), a medicinal mixture of Oryeongsan (Wulingsan) and Sagunza-tang (Sijunzi-tang) in Korean traditional medicine, against MCT-induced renal fibrosis and dysfunction. Although Oryeongsan has been studied for its effects on blood pressure regulation, improvement of physical and mental symptoms, and maintenance of fluid balance^{25,26,27} there is limited scientific evidence supporting the efficacy of GRT, which combines Oryeongsan and Sagunza-tang. Therefore, further research is needed to explore the effects of GRT. Additionally, there is a lack of objective pharmacological studies on the improvement of cardio-renal axis in PAH. Hence, this study aims to examine the effect of GRT on cardio-renal axis induced by MCT injection in rats.

2. Methods

2.1. Preparation of traditional Korean herbal formula: GRT decoction

GRT is composed of specific amounts of herbal ingredients as follows (Supplement 2): *Alisma orientale* Juzepzuk (93.75 g), *Glycyrrhiza uralen-*

sis Fisch. (46.875 g), *Polyporus umbellatus* Fries (56.25 g), *Cinnamomum cassia* J. Presl (18.75 g), *Atractylodes macrocephala* Koidzumi (56.25 g), *Panax ginseng* C. A. Meyer (46.875 g), *Poria cocos* Wolf (56.25 g), and *Poria cocos* Wolf (46.875 g). These herbal ingredients were obtained from the Herbal Medicine Cooperative Association (Iksan, Jeollabuk-do). To prepare the herbal formulation, the herbs were boiled in 2 liters of distilled water (DW) for 2 h. The resulting solution was then centrifuged at 3000 rpm for 10 min at 4 °C. The supernatant was concentrated using a Rotary vacuum evaporator (N-11, Tokyo Rikakikai, Tokyo, Japan) to obtain a powder. For in vivo experiments, this powder was dissolved in DW and appropriately diluted prior to oral administration. The dried GRT powder was stored at 4 °C until use.

2.2. High-performance liquid chromatography (HPLC) analysis

HPLC quantitative analyses were carried out using a SHIMADZU I-series LC-2030C instrument and a photodiode-array (PDA) detector (Shimadzu, Japan). Analysis of compounds were performed using a Capcell pak UG80 column, 4.6 mm I.D. x 250 mm, 5 μ m (Osaka Soda, Japan). The compounds used for analysis were cinnamic acid, glycyrrhizin, and ginsenoside Rb1 (Chemfaces, Wuhan, Hubei, China); Atractylenolide III (Merck, Rahway, NJ, USA) was purchased. The test solution was prepared three times and the average and standard deviation of the results were expressed. In addition to the quantified standards, several unidentified peaks were observed in the HPLC chromatogram. However, due to the complex nature of the multi-herbal formulation and the limited availability of reference compounds, these peaks were not further characterized in the present study.

2.3. Experimental animals

Male Sprague Dawley rats weighing between 165 and 185 g were procured from Koatech (Pyeongteak, Korea). The rats were housed under a 12-hour light and dark cycle. At day 0, the rats in the MCT, losartan (LOS), and GRT groups were subcutaneously injected with MCT (50 mg/kg, Sigma-Aldrich, Saint Louis, Missouri, USA) in MCT. The CON rats received a subcutaneous injection of saline only at day 0. From day 5 to day 20, rats in the CON and MCT groups were orally administered the vehicle daily. On the other hand, the LOS and GRT groups received LOS (20 mg/kg/day, TCI, Tokyo, Japan) and GRT (100 mg/kg/day) from day 5 to day 20, respectively. At day 20, animals from each group were individually housed in separate metabolic cages for one day. The rats were randomly divided into groups, with 10 rats in each group (Fig. 1A). Prior to euthanasia, the rats were anesthetized by inhalation of 4 % isoflurane liquid (Hana Pharm, Korea) mixed with oxygen in a chamber for 5–7 min, followed by cervical dislocation. This animal study was conducted with the approval of the Institutional Animal Care and Use Committee (IACUC) of Wonkwang University (WKU21–56).

2.4. PV loop assessment

Assessment of pressure-volume (PV) loop in ventricle was performed by PV loop system of Millar instruments (SPR-838, Houston, TX, USA). Rats were anesthetized with 4 % isoflurane inhalation mixed with oxygen and PV catheter was inserted into the RV through the right jugular vein.

2.5. Monitoring of renal function

Experimental animals were individually housed in metabolic cages for a duration of one day to allow for quantitative urine collections, as well as measurements of water and food intake. From day 20 at 09:00 a.m. to day 21 at 09:00 a.m., 24-hour urine samples were collected to determine the levels of creatinine (Cr), sodium, potassium, chloride,

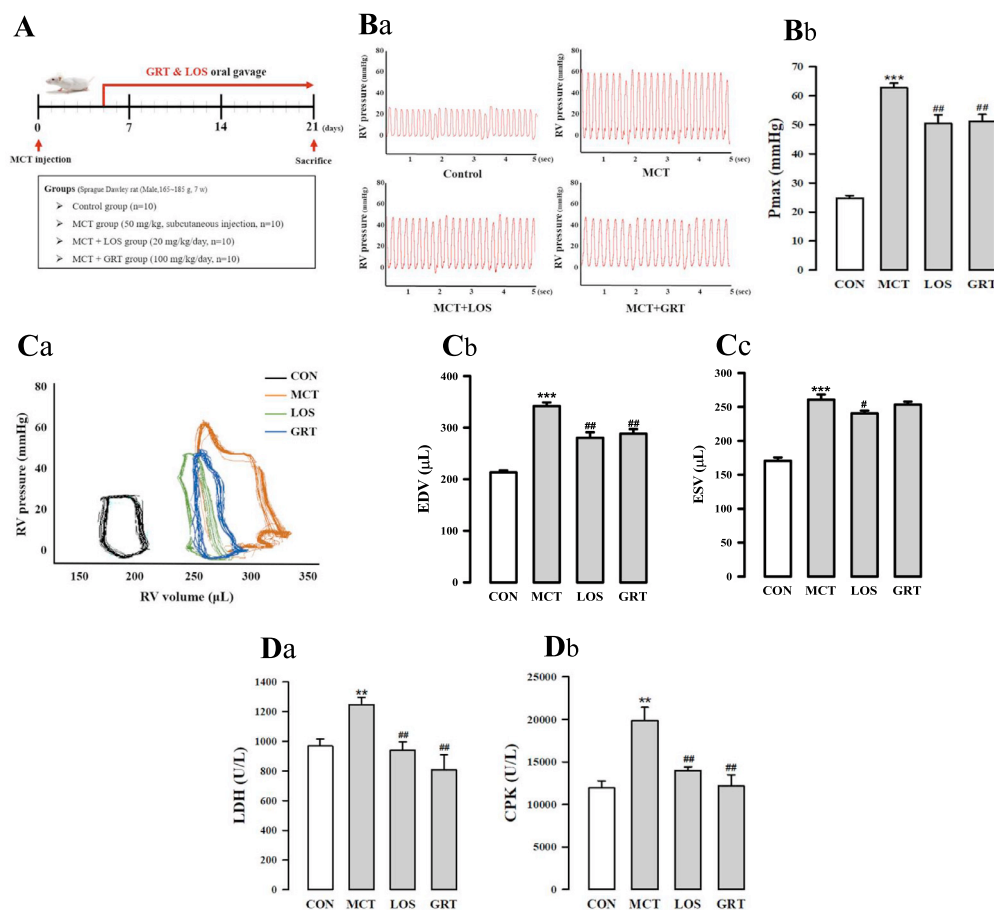


Fig. 1. Effect of GRT on RV pressure and pressure-volume loop in MCT-induced PAH-r. (A) The schematic diagram of an experiment to confirm the improvement effect of GRT by inducing MCT-induced PAH-r. (Ba) Representative physiological activity graph for RV pressure change. (Bb) Graph of changes in Pmax in MCT-induced PAH-r. (Ca) Representative physiological activity graph for RV pressure-volume change. (Cb) Graph of changes in EDV and (Cc) ESV in MCT-induced PAH-r. (Da) GRT reduces elevated plasma LDH and (Db) CPK in MCT-induced PAH-r. Values are expressed as mean $\pm$ S.E. ($n = 5$ per group). CON, control; CPK, creatine phosphokinase. EDV, end-diastolic volume; ESV, end-systolic volume; GRT, gunryeong-tang 100 mg/kg/day treated MCT-induced rats; LDH, lactate dehydrogenase; LOS, losartan 20 mg/kg/day treated MCT-induced rats; MCT, monocrotaline 50 mg/kg/day treated rats; PAH-r; pulmonary arterial hypertension rats; Pmax, maximum pressure; RV, right ventricle. $^{**}p < 0.01$ vs. CON; $^{##}p < 0.01$ vs. MCT.

and urinary osmolality (Uosmol). On the day of sacrifice, blood samples were obtained immediately following decapitation for the measurement of blood urea nitrogen (BUN), creatinine (Cr), lactate dehydrogenase (LDH), albumin, creatine phosphokinase (CPK), KIM-1, and NGAL. Plasma levels of BUN, Cr, LDH, albumin, and CPK were measured using an automated clinical chemistry analyzer (FUJI DRI-CHEM NX700, FUJIFILM Corporation, Tokyo, JAPAN). Plasma NGAL levels were measured using a kit from BioPorto (Gentofte, Denmark), while plasma levels of IL-6, TNF- α , IL-1 β , and KIM-1 were measured using a kit from R&D Systems (Minneapolis, MN, USA). Ion concentrations were determined using an electrolyte analyzer (NOVA 4, Biochemical, Waltham, MA, USA). Urinary osmolality (Uosmol) was measured using an Advanced CRYOMATICTM osmometer (Model 3900, Advanced Instruments Inc., Norwood, MS, USA). Urine Cr levels were determined via the colorimetric method using a spectrophotometer (Milton Roy, Rochester, NY, USA). Creatinine clearance (Ccr) values were calculated using the following formula: Ccr (ml/min/kg) = urine Cr (mg/ml) $\times$ UV (ml/kg/min) / plasma Cr (mg/ml).

2.6. Western blot analysis

Protein extraction was performed on RV tissue homogenate obtained from rats using a protein extraction solution (iNTRON Biotechnology, Korea). A total of 40 μ g of protein samples were separated through

10 % SDS-polyacrylamide gel electrophoresis and subsequently transferred onto a nitrocellulose membrane. The blots were washed using TBS-T (10 mM Tris-HCl, pH 7.6, 150 mM NaCl, 0.05 % Tween-20), followed by blocking with 5 % bovine serum albumin in TBS-T for 2 h. Next, the blots were incubated with specific primary antibodies at the recommended dilutions provided by Santa Cruz Biotechnology (Dallas, TX, USA). On the following day, a secondary antibody was applied and allowed to react for 1 hour. The protein bands were visualized using enhanced chemiluminescence (EzWestLumi plus, ATTO Technology, NY, USA). To determine protein expression levels, the signals captured on the nitrocellulose membranes were analyzed using the iBright FL100 image analyzer (Thermo Fisher Scientific, Waltham, MA, USA). The primary antibodies used in this study were obtained from Santa Cruz Biotechnology (Dallas, TX, USA) and included HMGB-1 (30 kDa), TLR4 (95/120 kDa), MyD88 (33 kDa), NF- κ B (65 kDa), p65, p-Smad 2 (60 kDa), Smad 2 (55~60 kDa), TGF- β (25 kDa), IL-6 (21~28 kDa), TNF- α (26 kDa), IL-1 β (17 kDa), KIM-1 (90~100 kDa), NGAL (25 kDa), glyceraldehyde-3-phosphate dehydrogenase (GAPDH, 37 kDa), and β -actin (42 kDa).

2.7. qRT-PCR

According to the manufacture's instructions, total RNA was extracted from RV tissue using Tri-RNA Reagent (FAVORGEN, Kaohsiung, Tai-

wan) and reverse transcribed into complementary deoxyribonucleic acid (cDNA) using a cDNA synthesis kit (Enzynomics, Korea). Each cDNA was amplified in a 20 μ l reaction system by using SYBR Green quantitative real time-polymerase chain reaction (qRT-PCR) Master Mix (Enzynomics, Korea) in 40 cycles of 95 °C and 60 °C (StepOnePlus qRT-PCR system, Applied Biosystems, Waltham, MA, USA). The primers were used for amplification (Supplement 3). Each gene expression was normalized to the endogenous α -tubulin. For each experimental condition, all qRT-PCRs were repeated at least three times.

2.8. Hematoxylin & eosin staining, picosirius red staining, masson trichrome staining

The kidneys, lungs, and hearts were isolated from each rat and fixed in 10 % formalin in 0.01 M phosphate-buffered saline for a duration of 7 days. Subsequently, the renal, lung, and cardiac tissues were embedded in paraffin, sectioned into 6 μ m slices, and placed on slides. Hematoxylin and eosin (H&E) stain was applied to the heart tissue slides for histopathological comparisons, and the slides were examined under a light microscope (EVOSTM M5000, Thermo Fisher Scientific, Bothell, WA, USA). Additionally, to assess cardiac fibrosis, the heart tissues were stained using the Masson trichrome stain kit (BBC Biochemical, USA). Histological changes were analyzed using the light microscope (EVOSTM M5000, Thermo Fisher Scientific, Bothell, WA, USA).

2.9. Statistical analysis

All experiments were conducted in triplicate, and statistical analyses were performed using the *t*-test. The results were presented as mean $\pm$ standard error (S.E.). The statistical significance between group means was determined using Student's *t*-test, with *p* < 0.05 considered as indicating a statistically significant difference.

3. Results

3.1. HPLC chromatograms of compounds from traditional Korean herbal formula (GRT)

HPLC analysis of cinnamic acid, glycyrrhizin, ginsenoside Rb1 and atractylenolide III for GRT, as shown in Supplement 1. As a result, 0.1 g of GRT contains 0.173 $\pm$ 0.001 mg of cinnamic acid; 17.689 $\pm$ 0.010 mg of glycyrrhizin; 0.701 $\pm$ 0.037 mg of ginsenoside Rb1; and 1.175 $\pm$ 0.013 mg of atractylenolide III. The molecular weights of these compounds range from 148.16 g/mol (cinnamic acid) to 1109.29 g/mol (ginsenoside Rb1), reflecting the considerable chemical diversity of the formulation. In addition to these four quantified marker compounds, the chromatogram showed several other peaks, which were not identified or quantified in this study. This limitation was due to the complexity of the formulation and the unavailability of appropriate reference standards.

3.2. Effect of GRT on RV pressure and pressure-volume loop in MCT-induced PAH-rats

The MCT is commonly used to make experimental PAH-rats (PAH-r). An apparent elevation of RV pressure occurred as a result of increased pulmonary arterial pressure in the MCT group. To investigate the effect of GRT on RV pressure, PV loop was performed. RV pressure was increased in rat with MCT, which was markedly restored by oral administration of LOS and GRT (Fig. 1Ba). Furthermore, maximum pressure (Pmax) level was increased by MCT injection, which was markedly restored by oral administration of LOS and GRT (Fig. 1Bb). Fig. 1Ca showed the representative PV loops for each group. The RV PV loop of MCT group was shifted to the right and pressure level was increased in MCT group. End-diastolic volume (EDV, Fig. 1Cb) and end-systolic volume (ESV, Fig. 1Cc) were significantly increased in MCT group. EDV

levels were remarkably restored by oral administration of LOS and GRT. In ESV level, there was no difference between MCT and GRT group (Fig. 1C). To investigate the effect of GRT on LDH, one of the hematological biomarkers of cardiac hypertrophy, hemodynamic analysis was performed (Fig. 1Da). Furthermore, high levels of CPK usually indicate some kind of stress or injury to the heart or other muscles. In this study, the plasma level of CPK was increased in MCT group, however, there was decreased in LOS and GRT group (Fig. 1Db).

3.3. Effects of GRT on lung morphology, fibrosis, and pulmonary vascular remodeling in MCT-induced PAH-r

Fig. 2A showed the representative image of whole lung in each group. Lung weights of all rats in MCT group were significantly increased and there was markedly decreased by oral administration of LOS and GRT (Fig. 2B). In lung weight/body weight (BW), lung/BW ratio was significantly increased in MCT group, which was markedly decreased by oral administration of LOS and GRT. To investigate the effect of GRT on lung fibrosis, IHC staining was performed (Fig. 2C). The protein expressions of fibrosis marker, such as collagen III, were increased in MCT group, however, these protein expressions were decreased in LOS and GRT group (Fig. 2D). To examine the impact of GRT on pulmonary artery fibrosis, we conducted Masson trichrome staining (Fig. 2E). The percentage of WT (%) was significantly elevated in the MCT group, but this increase was significantly attenuated by administering LOS and GRT orally (Fig. 2F). The percentage of WA (%) was significantly higher in the MCT group compared to the control, and there was no significant difference between the MCT group and the GRT group (Fig. 2G). The fibrosis area of the pulmonary artery was amplified in the MCT-induced pulmonary arterial hypertensive group. However, oral administration of LOS and GRT led to a marked reduction in the fibrosis area (Fig. 2H).

3.4. Effects of GRT on cardiac morphology in MCT-induced PAH-r

Fig. 3A is a representative heart photograph image for comparison of each group. First of all, when compared only with the difference in body weight of rats in each group, MCT-induced PAT-r showed a significant decrease compared to control, but it was not significantly different from GRT-treated (Fig. 3B). To confirm cardiac hypertrophy, the heart weight (HW)/BW ratio was calculated, and it was confirmed that GRT treatment significantly improved cardiac hypertrophy (Fig. 3C). In addition, right atrium (RA, Fig. 3D) and RV (Fig. 3E) hypertrophy were specifically observed in MCT-induced PAT-r, and it was confirmed that the difference was significantly reduced by GRT treatment when calculated as RV/LV+septum (Fig. 3G). In addition, hypertrophy of the right atrium (RA) and RV was specifically observed, and when the difference was calculated as RV/LV+septum, a significant difference by GRT could be compared. In addition, H&E staining was performed to investigate the effect of GRT on the diameter and thickness of RV when MCT-induced PAH-r (Fig. 3H). MCT-induced PAH-r resulted in cardiac hypertrophy and increased cardiomyocyte size, and treatment with GRT significantly reduced RV diameter (Fig. 3I) and thickness (Fig. 3J).

3.5. Effects of GRT on cardiac fibrosis and hypertrophy in MCT-induced PAH-r

To investigate the effect of GRT on RV fibrosis of cardiac tissue, masson trichrome staining and IHC staining were performed (Fig. 4). Although cardiac fibrosis area (%) was increased in MCT-induced PAH-r, it was markedly decreased by oral administration of LOS and GRT (Fig. 4A-B). Furthermore, collagen I, collagen III, and TGF- β expressions in MCT group rats with GRT was significantly decreased compared to MCT-induced PAH-r (Fig. 4C-E). In addition, to investigate the effect of GRT on cardiac hypertrophy, qRT-PCR was performed (Fig. 4F).

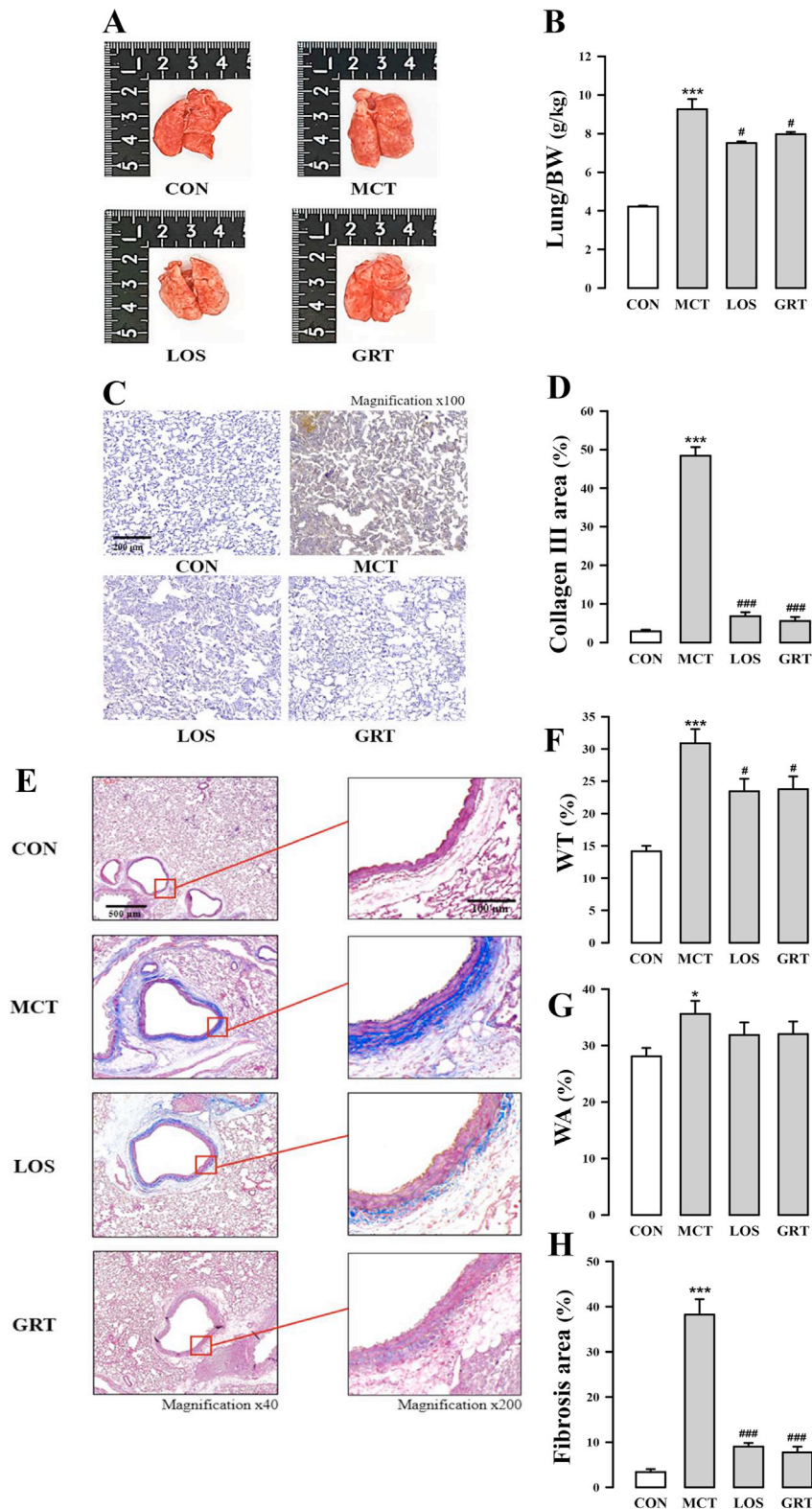


Fig. 2. Effects of GRT on lung morphology, fibrosis, and pulmonary vascular remodeling in MCT-induced PAH-r. (A) Representative images of lungs by group observed with the naked eye. (B) Graph of lung weight/body weight ratio change in MCT-induced PAH-r. (C) Representative images confirming collagen III expression by IHC staining to confirm the effect of GRT on improving pulmonary fibrosis in MCT-induced PAH-r. (D) Changes of collagen III area. (E) Pulmonary artery fibrosis assessed by Masson's trichrome staining in MCT-induced PAH-r (Magnification x40) and Histological changes confirmed by enlarging blood vessels in the stained image of the lungs (Magnification x200). (F) Changes in lung histology WT (%). (G) Changes in lung histology WA (%). (H) Changes in fibrosis area (%) in MCT-induced PAH-r. Values are expressed as mean $\pm$ S.E. ($n = 5$ per group). BW: body weight; CON, control; GRT, gunyeong-tang 100 mg/kg/day treated MCT-induced rats; IHC, Immunohistochemistry; LOS, losartan 20 mg/kg/day treated MCT-induced rats; MCT, monocrotaline 50 mg/kg/day treated rats; PAH-r; pulmonary arterial hypertension rats; WT (%), ratio of vascular wall thickness; WA (%), ratio of vascular wall area. * $p < 0.05$ and *** $p < 0.001$ vs. CON; # $p < 0.05$, ### $p < 0.01$, and #### $p < 0.001$ vs. MCT.

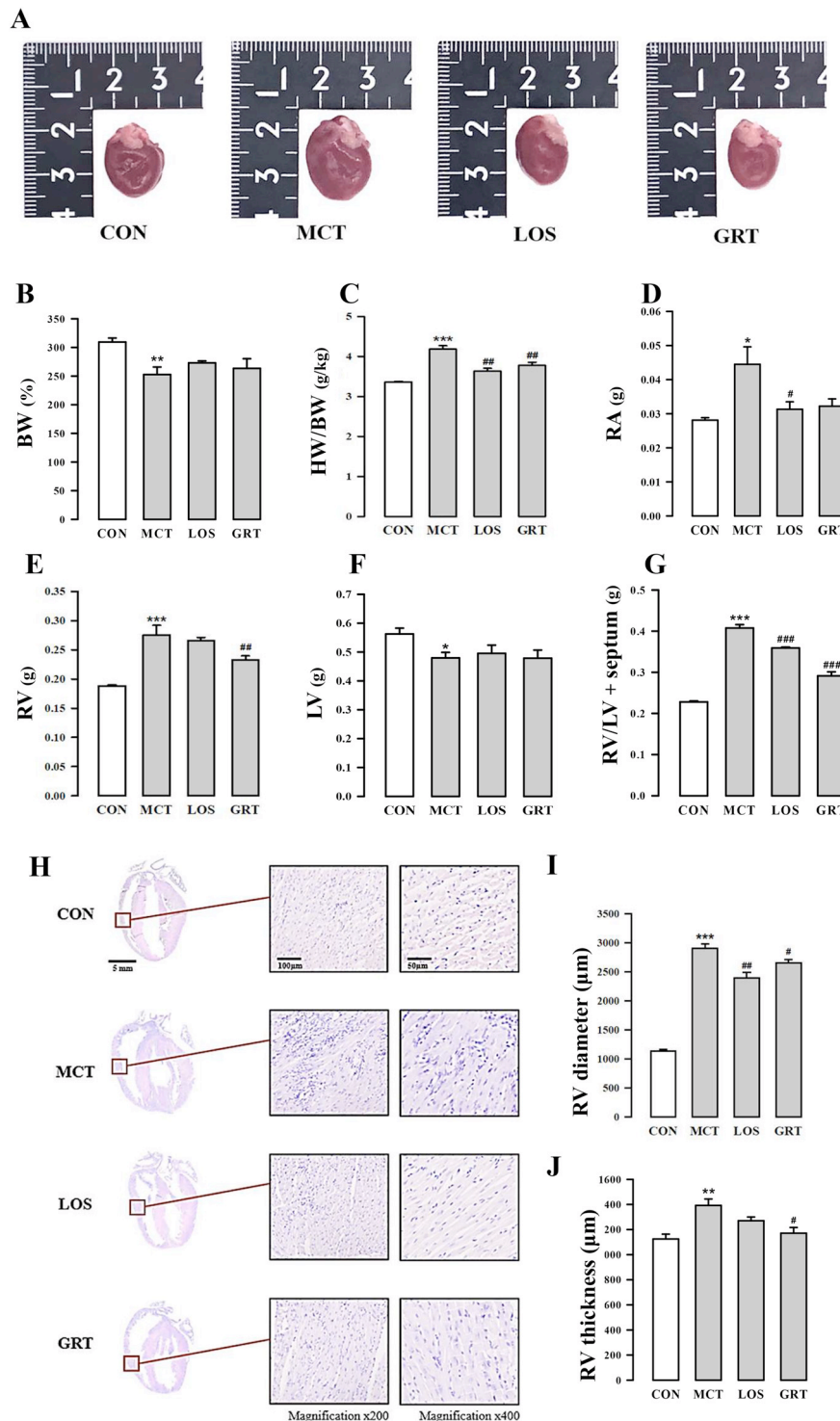


Fig. 3. Effects of GRT on cardiac morphology in MCT-induced PAH-r. (A) Representative images of hearts by group observed with the naked eye. Graphs of (B) BW, (C) HW/BW ratio, (D) RA, (E) RV, (F) LV, and (G) RV/LV+septum ratio change in MCT-induced PAH-r. RV hypertrophy was assessed by H&E staining in MCT-induced pulmonary arterial hypertension rats. (H) Histological changes were confirmed by enlarging myocardiocyte's size in the stained image of the heart (Magnification x200 & x400). (I) Changes in RV diameter in MCT-induced PAH-r. (J) Changes in RV thickness in MCT-induced PAH-r. Values are expressed as mean $\pm$ S.E. ($n = 5$ per group). BW: body weight; CON, control; GRT, gunryeong-tang 100 mg/kg/day treated MCT-induced rats; HW: heart weight; LOS, losartan 20 mg/kg/day treated MCT-induced rats; LV: left ventricle; MCT, monocrotaline 50 mg/kg/day treated rats; PAH-r; pulmonary arterial hypertension rats; RA: Right atrium; RV: right ventricle. * $p < 0.05$, ** $p < 0.01$, and *** $p < 0.001$ vs. CON; # $p < 0.05$, ## $p < 0.01$, and ### $p < 0.001$ vs. MCT.

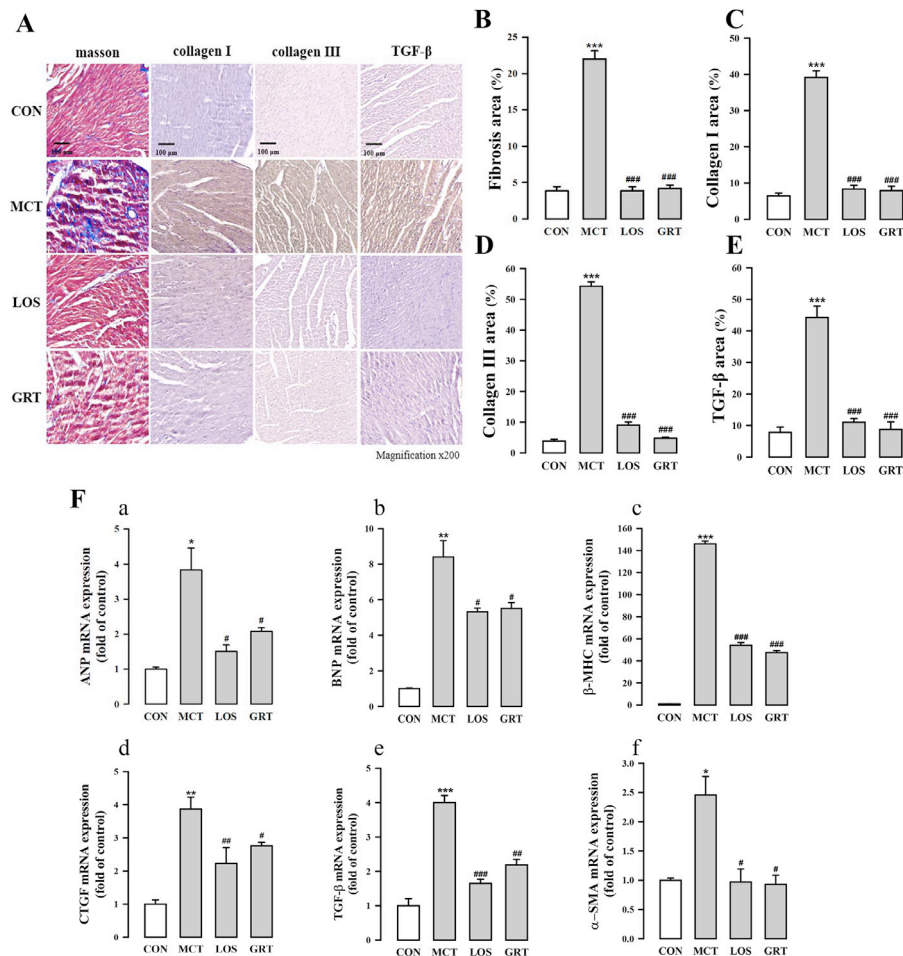


Fig. 4. Effects of GRT on cardiac fibrosis and hypertrophy in MCT-induced PAH-r. (A) Pathologic morphology and collagen I,III, and TGF- β deposition were assessed using masson trichrome staining and IHC staining (Magnification x200). Graphs of changes in (B) fibrosis area, (C) collagen I, (D) collagen III, and (E) TGF- β area in MCT-induced PAH-r. Effect of GRT on mRNA expression of cardiac hypertrophy in RV of MCT-induced PAH-r. Cardiac hypertrophy genes including (Fa) ANP, (Fb) BNP, and (Fc) β -MHC were measured by qRT-PCR. Cardiac fibrosis genes including (Fd) CTGF, (Fe) TGF- β , and (Ff) α -SMA were measured by qRT-PCR. Values are expressed as mean $\pm$ S.E. ($n = 5$ per group). α -SMA, α -smooth muscle actin; ANP, atrial natriuretic peptide; BNP, brain natriuretic peptide; β -MHC, β -myosin heavy chain; CTGF, connective tissue growth factor; CON, control; GRT, gunryeong-tang 100 mg/kg/day treated MCT-induced rats; LOS, losartan 20 mg/kg/day treated MCT-induced rats; MCT, monocrotaline 50 mg/kg/day treated rats; PAH-r; pulmonary arterial hypertension rats; RV: right ventricle; TGF- β , transforming growth factor- β . * $p < 0.05$, ** $p < 0.01$, and *** $p < 0.001$ vs. CON; # $p < 0.05$, ## $p < 0.01$, and ### $p < 0.001$ vs. MCT.

The mRNA expressions of hypertrophy marker, such as ANP, BNP, β -MHC, CTGF, TGF- β , and α -smooth muscle actin (α -SMA) were increased in MCT group, however, those were decreased in LOS and GRT-treated group.

3.6. Effect of GRT on cardiac inflammation in MCT-induced PAH-r

Cytokines such as TNF- α , IL-1 β , and IL-6 play important roles in cardiac inflammation (Fig. 5). To investigate the effect of GRT on plasma levels cytokine including TNF- α , IL-1 β , and IL-6, ELISA analysis was performed (Fig. 5A). The cytokine expression was increased in MCT group, but decreased in LOS and GRT group (Fig. 5Aa-c). Also, to investigate the effect of GRT on cytokine expression, western blot analysis was performed (Fig. 5B). The protein expression of cytokine, such as TNF- α , IL-1 β , and IL-6, were increased in MCT group, however, those were decreased in LOS and GRT group (Fig. 5Ba-d).

3.7. Mechanism of improvement of MCT-induced PAH-r by GRT

HMGB-1/TLR4/MyD88/NF- κ B pathway plays an important role in cardiac dysfunction. To investigate the effect of GRT on HMGB-1/TLR4/MyD88/NF- κ B pathway, western blot analysis was performed

(Fig. 6A). The protein expressions of HMGB-1/TLR4/MyD88/NF- κ B pathway were increased in MCT group, however, those were decreased in LOS and GRT group (Fig. 6Aa-e). Also, to investigate the effect of GRT on TGF- β /p-Smad 2 pathway, western blot analysis was performed (Fig. 6B). The protein expressions of TGF- β /p-Smad 2 pathway were increased in MCT group, however, those were decreased in LOS and GRT group (Fig. 6Ba-c).

3.8. Effects of GRT on renal function and fibrosis in MCT-induced PAH-r

To investigate the effect of GRT on UV, each group of rats was maintained in separate metabolic cage. UV was significantly decreased in MCT group, which was markedly increased by oral administration of GRT (Table 1). Uosmol was significantly decreased in MCT group, which was markedly restored by oral administration of GRT. These results indicated that PAH-r caused an impairment of urinary concentrating ability. Urinary sodium, potassium, and chloride excretion was decreased in rats with MCT, which was markedly restored by GRT treatment, respectively. Furthermore, Ccr was significantly decreased in MCT group compared with the CON group. The oral administration of GRT significantly restored Ccr level (Table 1). Renal dysfunction is typically characterized by an increase in KIM-1 and NGAL. To investigate the effect of GRT

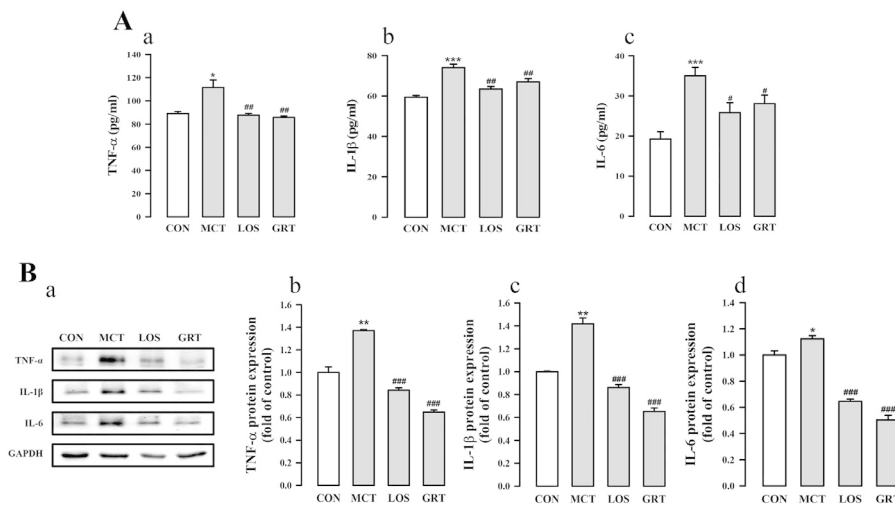


Fig. 5. Effect of GRT on cardiac inflammation in MCT-induced PAH-r. (A) Effect of GRT on plasma cytokine level and protein expressions of cardiac hypertrophy in RV of MCT-induced PAH-r. Plasma cytokine level including (Aa) TNF- α , (Ab) IL-1, and (Ac) IL-6 were analyzed by ELISA kit. (Ba) Protein including TNF- α , IL-1 β , and IL-6 were measured by western blot analysis. Graphs of changes in (Bb) TNF- α , (Bc) IL-1 β , and (Bd) IL-6 protein expressions in MCT-induced PAH-r. Values are expressed as mean $\pm$ S.E. ($n = 5$ per group). CON, control; GRT, gunyeong-tang 100 mg/kg/day treated MCT-induced rats; IL-1 β , interleukin-1 β (17 kDa); IL-6, interleukin-6 (21~28 kDa); LOS, losartan 20 mg/kg/day treated MCT-induced rats; MCT, monocrotaline 50 mg/kg/day treated rats; PAH-r, pulmonary arterial hypertension rats; TNF- α , tumor necrosis factor- α (26 kDa). * $p < 0.05$, ** $p < 0.01$, and *** $p < 0.001$ vs. CON; # $p < 0.05$, ## $p < 0.01$, and ### $p < 0.001$ vs. MCT.

Table 1

Effects of GRT on food and water intake, UV, Uosmol, urinary Na⁺, K⁺, Cl⁻ excretion, and Ccr.

	CON	MCT	LOS	GRT
Food intake (g)	23.0 $\pm$ 0.6	14.6 $\pm$ 1.3***	19.5 $\pm$ 1.7	19.4 $\pm$ 1.0#
Water intake (g)	32.6 $\pm$ 0.7	18.4 $\pm$ 2.4***	23.4 $\pm$ 1.8	25.2 $\pm$ 3.8
UV (ml/day)	11.7 $\pm$ 0.6	7.9 $\pm$ 0.8**	10.9 $\pm$ 0.6#	10.7 $\pm$ 0.9#
Uosmol (mOsm)	2219.4 $\pm$ 45.2	2016.6 $\pm$ 41.3*	2203.2 $\pm$ 66.7#	2128.8 $\pm$ 21.8#
UNaV (mmol/L)	181.8 $\pm$ 2.0	121.8 $\pm$ 7.7***	154.8 $\pm$ 9.4#	159.6 $\pm$ 7.9##
UKV (mmol/L)	275.2 $\pm$ 3.2	247.3 $\pm$ 4.0***	279.2 $\pm$ 8.4##	278.2 $\pm$ 6.9##
UClV (mmol/L)	280.8 $\pm$ 2.8	229.8 $\pm$ 7.8***	271.2 $\pm$ 13.4#	257.4 $\pm$ 7.1#
Urinary Cr (mg/ml)	17.9 $\pm$ 0.8	18.5 $\pm$ 1.2	18.3 $\pm$ 1.0	16.4 $\pm$ 0.8
Ccr (ml/min/mg)	0.0183 $\pm$ 0.0007	0.0116 $\pm$ 0.0003***	0.0175 $\pm$ 0.0009###	0.0174 $\pm$ 0.0003###

Values are expressed as mean $\pm$ S.E. ($n = 5$ per group). Ccr, creatinine clearance; CON, control; Cr, creatinine; LOS, losartan 20 mg/kg/day treated MCT-induced rats; GRT, gunyeong-tang 100 mg/kg/day treated MCT-induced rats; MCT, monocrotaline 50 mg/kg/day treated rats; UV, urinary volume; Uosmol, urinary osmolality; UNaV, urinary excretion of Na⁺; UKV, urinary excretion of K⁺; UClV, urinary excretion of Cl⁻. * $p < 0.05$, ** $p < 0.01$, and *** $p < 0.001$ vs. CON; # $p < 0.05$, ## $p < 0.01$, and ### $p < 0.001$ vs. MCT.

on renal dysfunction, plasma levels (Fig. 7A) and western blot analysis (Fig. 7B) were performed. The results show that KIM-1 and NGAL were increased in MCT group. However, GRT treatment downregulated KIM-1 and NGAL, respectively. Thus, GRT suppressed MCT-induced renal dysfunction in PAH-r. Furthermore, BUN levels were also increased in plasma MCT group. However, GRT treatment significantly decreased the BUN level (Fig. 7Ac). Plasma Cr was increased in MCT-induced PAH-r, however, there was no significant change in GRT group (Fig. 7Ad). Picrosirius red staining was performed to examine the impact of GRT on renal fibrosis, as depicted in Fig. 7C. Administration of MCT led to the development of renal fibrosis in the MCT group, whereas treatment with GRT exhibited a protective effect against renal fibrosis in the renal cortex, outer medulla, and inner medulla. Additionally, GRT treatment demonstrated the ability to preserve tubular integrity and prevent tubular lesions induced by MCT injection. These findings suggest that GRT has the potential to mitigate pathological kidney damage associated with renal dysfunction (Fig. 7).

4. Discussion

GRT was documented in a traditional Chinese medical book and consists of 8 herbal medicines: *Alisma orientale* Juzepzuk, *Glycyrrhiza uralensis* Fisch., *Polyporus umbellatus* Fries, *Cinnamomum cassia* J. Presl,

Atractylodes macrocephala Koidzumi, *Panax ginseng* C. A. Meyer, and *Poria cocos* Wolf. GRT has been traditionally used to treat edema induced by yin deficiency.

GRT, composed of Oryeongsan and Sijunzi-tang, is expected to exert a synergistic therapeutic effect by combining the diuretic and RAAS-inhibitory properties of Oryeongsan with the gastrointestinal and energy-metabolism enhancing actions of Sijunzi-tang, which may also improve the bioavailability of co-administered components.^{2,3,28,29} However, there is no evidence supporting the effectiveness of GRT in MCT-induced cardio-renal axis associated with PAH-r. Therefore, the present study aimed to investigate whether GRT improves cardio-renal function in rats with MCT-induced PAH-r. The study revealed that GRT effectively ameliorated elevated RV systolic pressure and attenuated RV hypertrophy in rats with MCT-induced PAH-r. Administration of GRT not only attenuated RV hypertrophy but also mitigated MCT-induced pulmonary vascular remodeling. The study observed a significant increase in ANP, BNP, β -MHC expressions, collagen I and III deposition resulting from MCT injection, which were significantly improved by GRT treatment. Furthermore, damage to the heart muscle tissue leads to the leakage of CPK into the blood, commonly indicating heart or muscle damage.^{30,31} Additionally, the study demonstrated various changes in cardiac biomarkers, including significant increases in CPK and LDH compared to the control group, which were previously reported to be

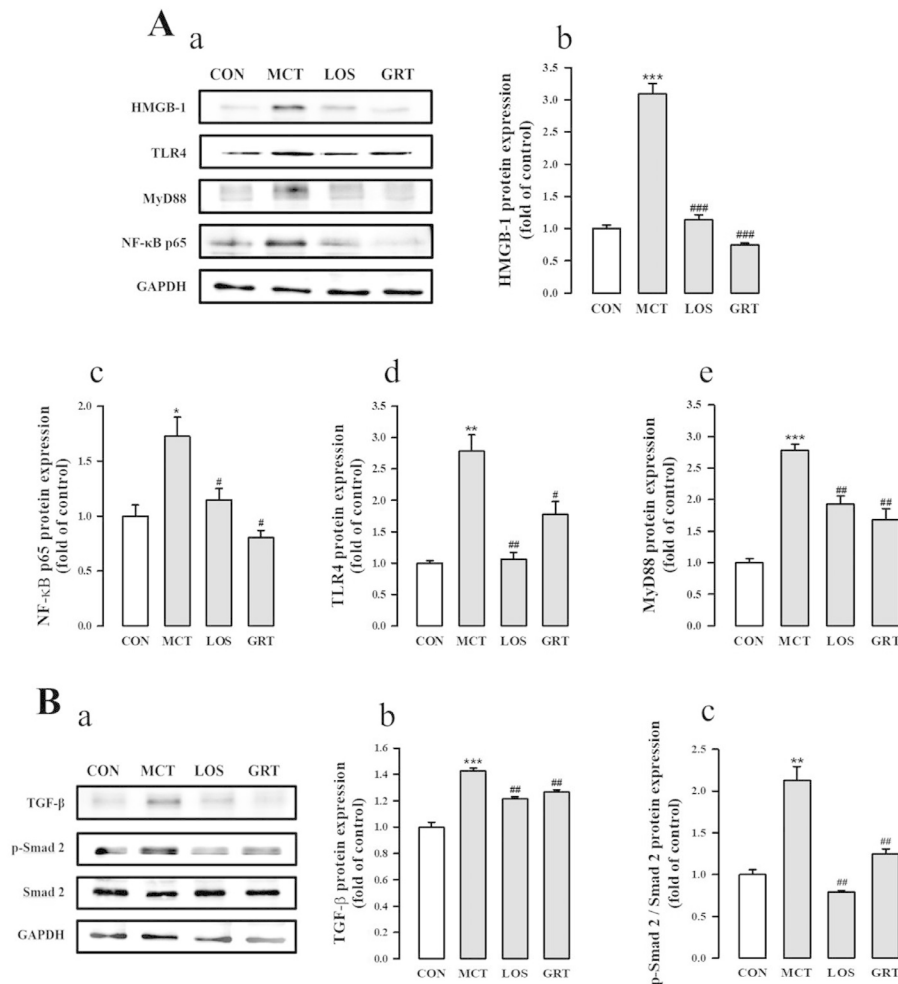


Fig. 6. Effects of GRT on protein expressions of inflammasomes pathway in RV of MCT-induced PAH-r. (Aa) Protein including HMGB-1, TLR4, MyD88, and NF-κB pathway were measured by western blot analysis. Graphs of changes in (Ab) HMGB-1, (Ac) NF-κB, (Ad) TLR4, and (Ae) MyD88 protein expressions in MCT-induced PAH-r. (Ba) Effects of GRT on protein expressions of TGF-β/p-Smad 2 pathway in RV of MCT-induced pulmonary arterial hypertensive rats. Graphs of changes in (Bb) TGF-β and (Bc) p-Smad 2/Smad 2 protein expressions in MCT-induced PAH-r. Values are expressed as mean ± S.E. ($n = 3$ per group). CON, control; GRT, gunryeong-tang 100 mg/kg/day treated MCT-induced rats; HMGB-1, high mobility group box-1 (30 kDa); LOS, losartan 20 mg/kg/day treated MCT-induced rats; MCT, monocrotaline 50 mg/kg/day treated rats; MyD88, myeloid differentiation primary response 88 (33 kDa); NF-κB, nuclear factor-kappa B (65 kDa); PAH-r, pulmonary arterial hypertension rats; p-Smad-2, phospho-smad-2 (60 kDa); Smad-2, mothers against decapentaplegic homolog 2 (55~60 kDa); TGF-β, transforming growth factor-β (25 kDa); TLR4, toll-like receptor 4 (95/120 kDa). * $p < 0.05$, ** $p < 0.01$, and *** $p < 0.001$ vs. CON; # $p < 0.05$, ## $p < 0.01$, and ### $p < 0.001$ vs. MCT.

associated with cardiac dysfunction.³¹ Overall, GRT exhibited a noticeable effect on reversing cardiac fibrosis, suggesting its potential as a novel treatment for PAH-r.

Previous studies have established that inflammation plays a role in the pathogenesis of PAH, and perivascular inflammation is associated with vascular remodeling and increased pulmonary arterial pressure.³² Inflammation is triggered by the increase and activation of professional inflammatory cells induced by growth factors and cytokines. Inflammation-related factors are typically found in the lungs of patients with PAH, and plasma levels of pro-inflammatory cytokines such as TNF-α, IL-1β, and IL-6 are increased in MCT-treated rats.^{32,33} Elevated levels of TNF-α, IL-1β, and IL-6 in MCT-induced PAH-r suggest that increased cytokine levels may impact the survival rate of PAH.^{34,35} Moreover, TNF-α, IL-1β, and IL-6 contribute to the pathogenesis of PAH.^{36,37} The study demonstrated that MCT injection elevated plasma levels of pro-inflammatory cytokines, which were reversed by GRT treatment. TGF-β regulates various signaling pathways that promote smooth muscle cell proliferation, differentiation, and endothelial interstitial transformation, ultimately leading to the pathogenesis of PAH.³⁸ Based on these findings, TGF-β is inferred to play an important role in the devel-

opment of MCT-induced PAH-r. The present study found significantly elevated TGF-β protein and mRNA levels in rats with MCT-induced PAH-r, whereas GRT treatment decreased TGF-β expression. MCT injection promotes the translocation of HMGB-1 from the nucleus to the cytoplasm and its extracellular release by inducing the lysine acetylation of HMGB-1. In this case, HMGB-1 can bind to TLR4 and activate MyD88 and NF-κB signaling pathways, ultimately leading to inflammation.³⁹ NF-κB serves as a central regulator of target genes involved in inflammation, vascular endothelial dysfunction, and angiogenesis.⁴⁰ The activation of the NF-κB pathway results in the expression of various genes encoding pro-inflammatory cytokines such as IL-1β and IL-6. Numerous studies have demonstrated the significant role of NF-κB in PAH-r. NF-κB activation has been observed in pulmonary arterial smooth muscle cells and macrophages derived from patients with idiopathic PAH, and NF-κB inhibition has shown improvement in MCT-induced PAH-r.^{40,41} In this study, GRT treatment reduced NF-κB p65 expression in the right ventricle of rats with pulmonary arterial hypertension. The study revealed that renal function, including Ccr, Uosmol, and urinary Na⁺, K⁺, and Cl⁻ excretion, was significantly decreased in PAH-r, indicating impaired urinary concentrating ability. However, GRT treatment improved renal

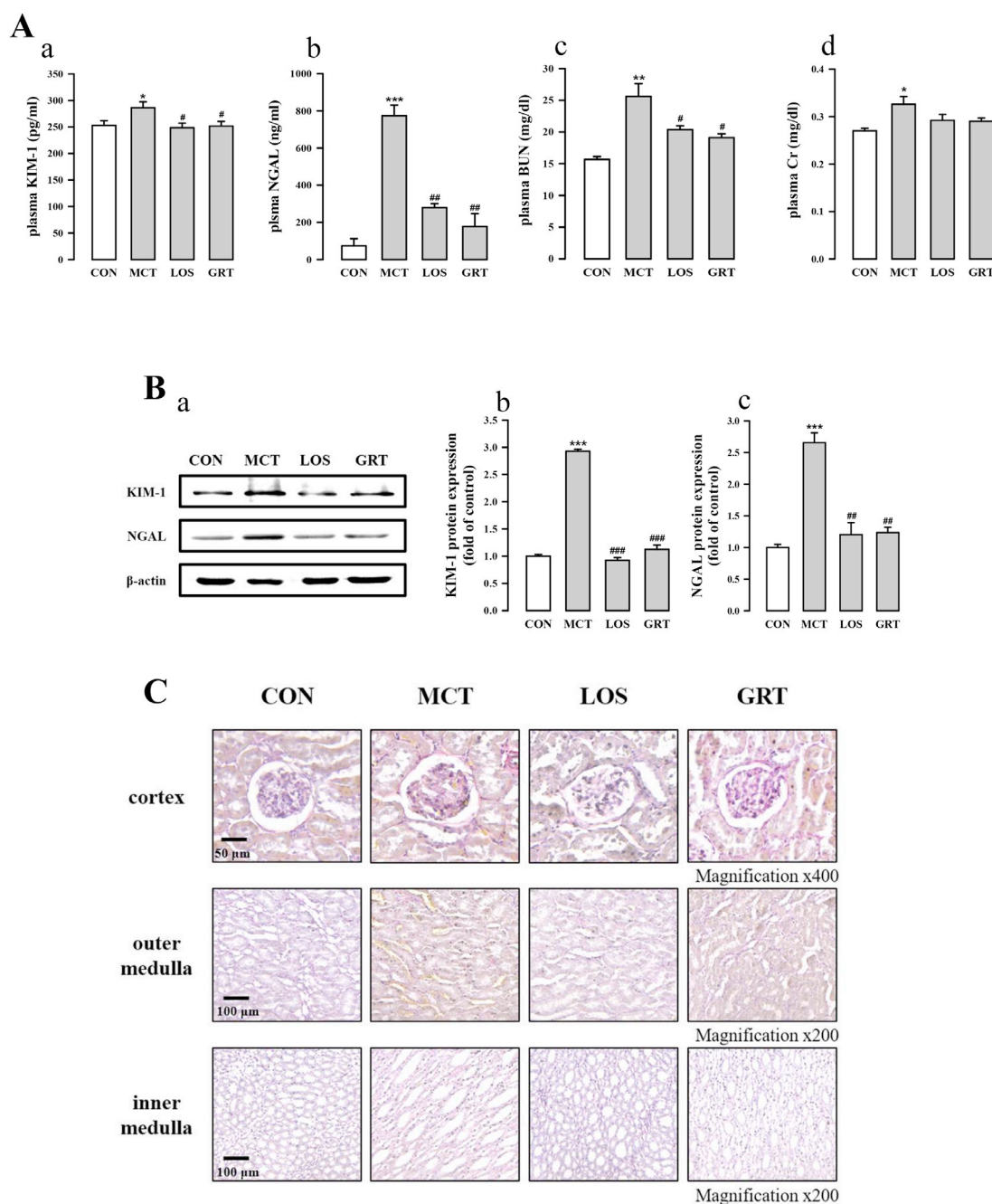


Fig. 7. Effects of GRT on renal function and fibrosis in MCT-induced PAH-r. Graph of changes in (Aa) plasma KIM-1, (Ab) NGAL, (Ac) BUN, and (Ad) Cr level in MCT-induced PAH-r. (Ba) Renal dysfunction protein including KIM-1 and NGAL were measured by western blot analysis. Graphs of changes in (Bb) KIM-1 and (Bc) NGAL protein expressions in MCT-induced PAH-r. Representative images of renal fibrosis MCT-induced PAH-r. Pathologic morphology and collagen deposition were assessed using picrosirius red staining (Magnification x400 & x200). Values are expressed as mean $\pm$ S.E. ($n = 3-5$ per group). β -actin, beta actin (42 kDa); CON, control; GRT, gunyeong-tang 100 mg/kg/day treated MCT-induced rats; KIM-1, kidney injury molecule-1(90–100 kDa); LOS, losartan 20 mg/kg/day treated MCT-induced rats; MCT, monocrotaline 50 mg/kg/day treated rats; NGAL, neutrophil gelatinase associated lipocalin (25 kDa); PAH-r; pulmonary arterial hypertension rats. *** $p < 0.001$ vs. CON; ## $p < 0.01$ and ### $p < 0.001$ vs. MCT.

function by enhancing Ccr, Uosmol, urinary volume (UV), and urinary Na^+ , K^+ , and Cl^- excretion in rats with MCT-induced PAH-r. Therefore, GRT was found to suppress renal dysfunction in rats with MCT-induced PAH-r.

In previous studies, it has been established that models of renal dysfunction exhibit tubular dilation, tubular epithelial injury, cast formation, and debris accumulation in the outer and inner medulla.^{42,43} Similarly, this study observed damage to the outer and inner medulla, as

well as the cortex, in rats with MCT-induced PAH-r. However, GRT treatment ameliorated renal injury by reducing tubular dilation, tubular epithelial injury, cast formation, and debris accumulation in rats with MCT-induced renal dysfunction. Furthermore, the study identified glomerular damage and vacuole formation in the renal cortex of rats with pulmonary arterial hypertension. These results demonstrated that GRT improved glomerular damage by suppressing vacuole formation in MCT-induced pulmonary arterial hypertensive rats. Previous studies

have shown that elevated levels of BUN, KIM-1, and NGAL are characteristic of PAH-r.^{44,45} The results of the present study were consistent with those of previous studies, indicating that renal functions, including BUN, KIM-1, and NGAL, were decreased in PAH-r. This study aimed to investigate whether GRT treatment inhibits renal failure. In conclusion, this study demonstrates that GRT ameliorates renal dysfunction, including BUN, KIM-1, and NGAL, in PAH-r.

Strengths of this study include its comprehensive evaluation of Gunryeong-tang (GRT) as a potential treatment for cardio-renal dysfunction induced by PAH-r. Additionally, previous studies have suggested that modulating the heart-kidney axis could be beneficial in cardio-renal syndrome.⁴⁶ The researchers examined multiple parameters, such as RV enlargement, fibrosis, lung weight, pulmonary fibrosis, and cardiac hypertrophy markers, providing a thorough understanding of GRT's effects. The use of appropriate animal models and diverse outcome measures enhances the reliability of the findings. Mechanistic insights were gained by investigating specific pathways, including the TGF- β /p-Smad2 and HMGB-1/TLR4/MyD88/NF- κ B pathways, which contributes to a deeper understanding of GRT's therapeutic effects. However, there are limitations to consider. The findings may have limited generalizability to human patients since animal models were used. The study did not compare GRT with other treatment options, which limits the assessment of its relative efficacy. Additionally, the focus on specific pathways and biomarkers may overlook other important molecular mechanisms involved in cardio-renal axis. The long-term effects and potential side effects of GRT treatment were not extensively investigated.

Although this study demonstrated the therapeutic potential of GRT in a PAH-induced cardio-renal dysfunction model, several limitations should be acknowledged. We performed HPLC-based quantification of four marker compounds (cinnamic acid, glycyrrhizin, ginsenoside Rb1, and atractylenolide III) to support formulation quality control. These compounds exhibited a wide range of molecular weights (148.16–1109.29 g/mol), reflecting the chemical heterogeneity of GRT. However, this targeted analysis provides only a partial view of the complex composition of the formulation. Several additional peaks were observed in the chromatogram, but we were unable to identify or characterize them due to the lack of reference standards and analytical constraints. This limited phytochemical profiling restricts our understanding of which constituents are responsible for the observed therapeutic effects. Moreover, the study did not include structural or confirmatory analyses using advanced techniques such as LC-MS/MS or NMR, which would enable more detailed molecular characterization. Future studies employing such methods are planned to establish a comprehensive chemical fingerprint of GRT. The selected dose of 100 mg/kg/day was based on preliminary efficacy and safety data. However, dose-response studies and long-term toxicity evaluations were not performed, which are essential steps for future preclinical development. In addition, tissue distribution and pharmacokinetics of the active constituents were not assessed. Considering the multi-component nature of the formulation, it is likely that individual compounds contribute to therapeutic effects through distinct mechanisms. Therefore, accurate quantification and standardization of these compounds will be crucial for advancing GRT as a clinically applicable treatment. These limitations were recognized based on the reviewer's valuable feedback, and we plan to address them in future investigations.

In conclusion, this study demonstrates the therapeutic potential of GRT in addressing cardio-renal axis associated with PAH-r. The findings suggest that GRT treatment exerts protective effects on cardio-renal dysfunction by inhibiting the TGF- β /p-Smad2 and HMGB-1/TLR4/MyD88/NF- κ B pathways. However, further research is needed to validate these findings in clinical settings, compare GRT's effectiveness with other treatments, explore additional molecular mechanisms, and assess its long-term efficacy and safety. Overall, GRT shows promise as an effective medicine for the treatment of cardio-renal axis associated with PAH-r, but more investigation is required for a comprehensive understanding of its potential benefits and limitations.

CRediT authorship contribution statement

Se Won Na: Methodology, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization. **Hye Yoom Kim:** Methodology, Formal analysis, Data curation, Writing – original draft, Visualization, Project administration, Funding acquisition. **Jung Joo Yoon:** Data curation, Writing – original draft. **Mi Hyeon Hong:** Investigation. **Ai Lin Tai:** Validation. **Yun Jeong Yang:** Validation, Formal analysis. **Chae Yeon Yoo:** Validation, Formal analysis. **Ho Sub Lee:** Conceptualization, Writing – review & editing, Supervision, Project administration. **Dae Gill Kang:** Conceptualization, Writing – review & editing, Supervision, Project administration.

Declaration of competing interest

All authors declare that there are no conflicts of interest.

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Ethics statement

This study followed the institutional ethics procedure for animal experiments.

Data availability

The data sets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.imr.2025.101218.

Supplement 1. HPLC Chromatograms of compounds from GRT.

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